Nile University
جامعة النيل الأهلية
RESEARCH OFFICE

CSCI417

NU RESEARCH FORUM 17TH UGRF

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Dysarthria Detection Using Machine Learning Techniques

ABSTRACT

Dysarthria, characterized by impaired muscle control in speech, presents a challenge in understanding varying degrees of severity among affected individuals. Our project focuses on employing machine learning models—specifically CNN, SVM, Decision Tree, Logistic Regression, and RNN—to detect dysarthria in individuals based on speech patterns. It aims to bridge this gap by exploring these models' effectiveness in discerning dysarthric speech.

INTRODUCTION

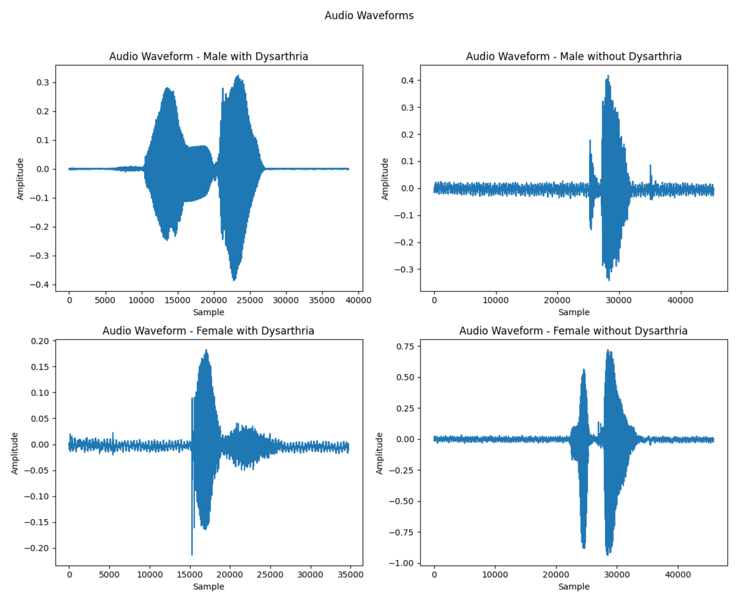
Dysarthria is a physiological disorder that affects speech by impairing coordinated muscular movements, which can be caused by weakness, paralysis, or sensory loss. Vital speech components such as respiration, phonation, resonance, articulation, and prosody are all impacted by this neurogenic speech problem, which is brought on by nervous system disorders. There are a variety of signs, such as fast and ambiguous speech, nasal or strained voice, and slurred, slow, or loud speaking. Within the domain of speech disorders, dysarthria emerges as a distinctive challenge. When it comes to speech characteristics, dysarthria can present in a variety of ways and to different degrees depending on the individual. Using machine learning techniques like Convolutional Neural Networks (CNN), Support Vector Machines (SVM), Decision Trees, Logistic Regression, and Recurrent Neural Networks (RNN), this project takes an approach to understanding dysarthric speech patterns.

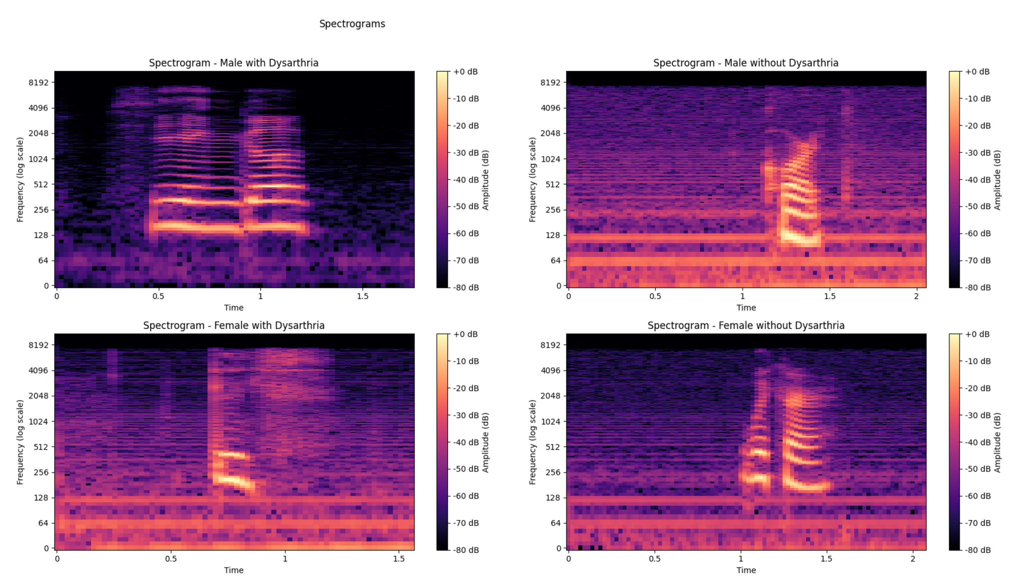
METHODOLOGY

The audio files recorded during clinical sessions, each associated with individual data such as gender, were organized in a CSV file, with entries indicating the gender and corresponding audio file names for each session participant.

- **Exploratory Data Analysis:**

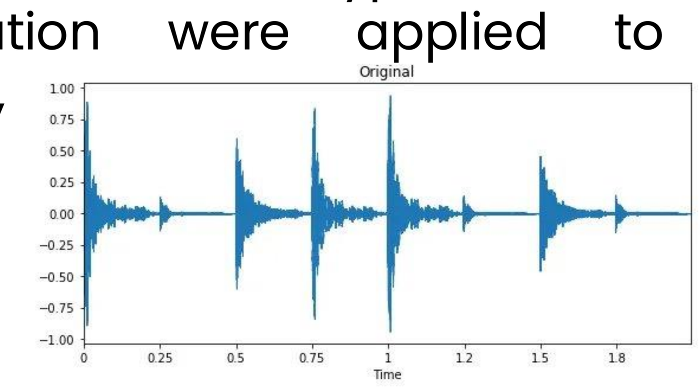
A comparison between visualizations was made for four random samples (Male, Female, dysarthric, non-dysarthric) like Waveplot, spectrogram, Spectral Rolloff and zero crossing rate to inspect the difference Between the dysarthric and the non-dysarthric.

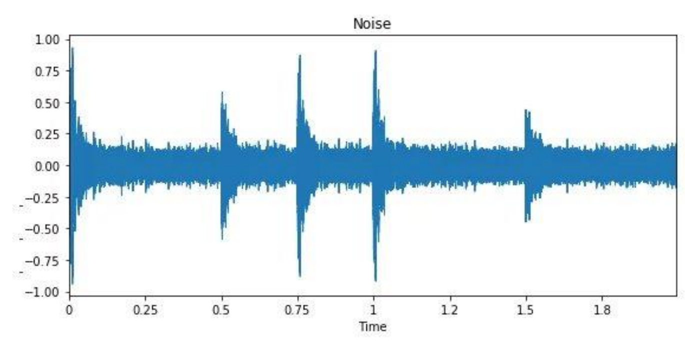




- **Data Augmentation:**

Although the dataset was originally augmented, two types of data augmentation were applied to the audio files, noise injection and time shifting to add noise to both classes to be Less prone for audio noise misclassification, and time shifting to detect whether the person is dysarthric or not from any start time.





- **Feature Extraction:**

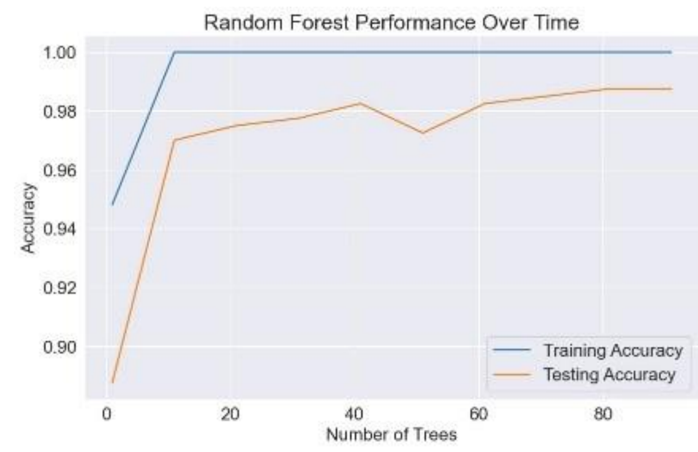
In the final preprocessing stage, the audio signals were transformed for a more structured format. Mel-frequency cepstral coefficients (MFCC) were employed to extract 39 numerical coefficients from the frequency domain of each audio signal, optimizing the dataset for model training.

- **Training:**

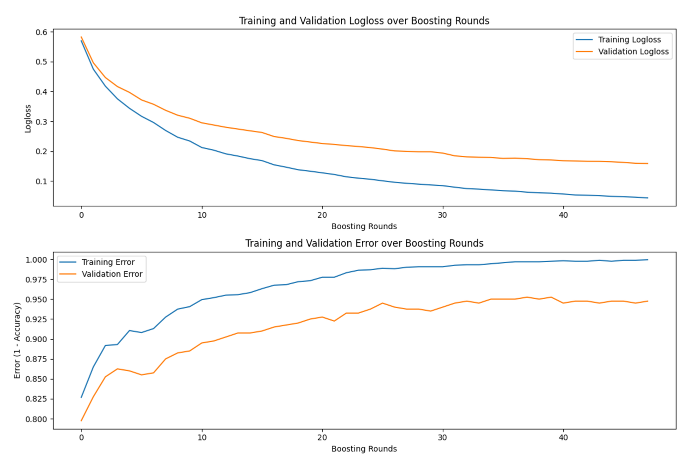
Lastly, 5 different models were implemented (SVM, Random Forest, Logistic Regression, RNN, XGB). Meanwhile, hyperparameter tuning was performed to identify the optimal parameters for each model.

RESULTS

The Random Forest model performs well on audio files due to its ensemble approach, combining multiple decision trees to enhance accuracy and stability. This model handled the non-linear relationship in the data.



The XGBoost model effectively classified dysarthria with a 0.96 cross-validation score and 0.9525 accuracy, indicating its strong capability in detecting speech Variations.



The rbf kernel of the support vector machine has achieved an accuracy of 97.75% while the linear kernel achieved 91% which indicates the data is not linearly separable. The rbf kernel has achieved an average cross validation score of 96.79%

```
from sklearn.model_selection import KFold, cross_val_score
k_folds = KFold(n_splits = 10)
scores = cross_val_score(clf, X, y, cv = k_folds)
print("Model: Support Vector Machine")
print("Cross Validation Scores: ", scores)
print("Average CV Score: ", scores.mean())
print("Number of CV Scores used in Average: ", len(scores))
```

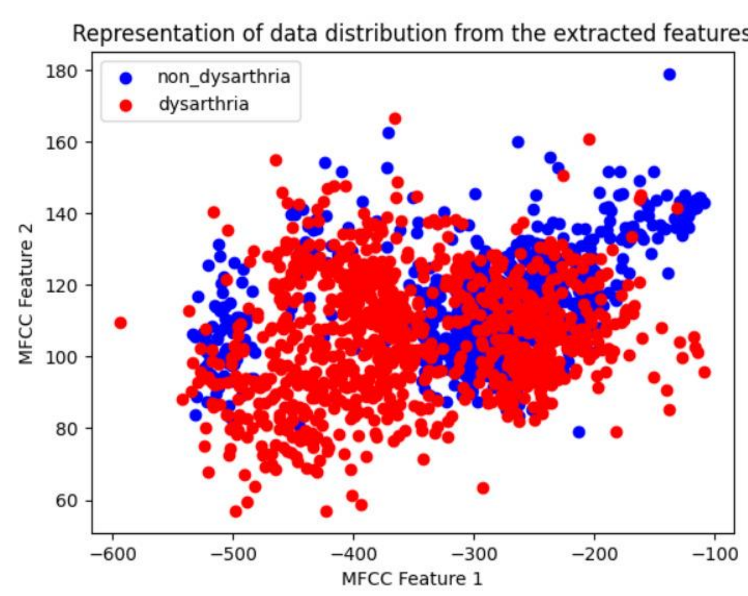
```
Model: Support Vector Machine
Cross Validation Scores:  [0.965      0.98       0.975      0.955      0.975      0.97
 0.95       0.97       0.975      0.96482412]
Average CV Score:  0.9679824120603016
Number of CV Scores used in Average:  10
```

```
from sklearn.metrics import accuracy_score
predictions = clf.predict(X_test)
print(accuracy_score(y_test, predictions))
```

```
0.9775
```

DISCUSSION & CONCLUSION

After assessing the five models on +90% accuracy because of the non-linear nature of the MFCC features. The image of reduced MFCC dimensions shows how a rbf kernel of SVM can perform better than the linear one. while a logistic regression model can struggle with that kind of data distribution. In summary, SVM, XGBoost, and Random Forest proved their efficiency for non-linear MFCC data.



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